

Building Environmental Maps of Human Activity for a Mobile Service Robot at the "Miraikan" Museum

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Abstract. This paper describes environment maps that are comprised of the following three types of information, 1) 3D environmental changes that represents human activities, 2) human trajectories in 2D that represent how humans move in the environment, and 3) human posture data. These maps are utilized in order to plan safer, quicker and/or non-human-disturbing paths for a mobile service robot at the museum "Miraikan". Experiments are conducted within "Miraikan" and results are shown.

1 Introduction

Autonomous carrier, cleaning, and security guard robots have recently become commercially available. The mobility function of such robots assumes quasi-static, indoor environments with few human inhabitants. Research in simultaneous localization and mapping (SLAM), mapping, localization, and path planning fields actively targets quasi-static environments. However, robots in human rich environments need further information about how humans use the environment to achieve safe and efficient behavior. Konolige et al. proposed lifelong environmental maps for

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daily living environments [1] to tackle this kind of problem that focused on temporal changes in map occupancy. Hamada et al. also proposed a human trajectory map that categorized human use by using moving speed and direction [2]. This information was used to bias results from robot path planning so that they could efficiently move through space cohabited by human beings.

We focused on a map used by a mobile service robot at a museum. Several such studies have been reported such as the RHINO robot at the Deutsches Museum in Bonn [3], and Minerva at the Smithsonian's National Museum of American History [4]. The Minerva study demonstrated that robots in museum situations sometimes encounter crowds of people, and not just single individuals, and the typical interaction time is quite short.

We have also done studies at the "Miraikan" Museum and identified three difficulties with mobility such as a) crowds of people moving in corridors and stopping in front of displays, b) frequently changing environments (such as fences, stools for small children, and guideboards), and c) changes in displays or exhibition areas. It is important for robots operating in museums, whatever their tasks are (such as guiding, giving tours, or cleaning), to better understand how humans use the environment. Therefore, we propose three types of maps to help robots understand the way humans use the environment: 1) a 3D environmental change map that represents human activities, 2) a map of human trajectories in 2D that represents how humans move in the environment, and 3) a human posture map to detect human interaction with museum displays.

2 Concept of Human Environment Maps

Maps are searched in the path planning phase for minimum cost paths by minimizing the accumulated costs that are assigned to each map cell. Although we can obtain minimum length paths, such a strategy might not be appropriate for museum environments. What we would like to obtain is somewhat safe, efficient, and nonhuman-disturbing paths. Three maps were constructed to obtain such paths: 1) a 3D environmental change map, 2) a map of human trajectories in 2D, and 3) a human posture map. The robot could understand more about the environment by combining these three maps and generate better paths to reach its goal. This information was also useful for museum management purposes, in addition to such direct usage by the robot, such as how many people looked at displays, how long they spent looking at them, the human density of corridors, and the existence of bottlenecks.

3 3D Environmental Change Map

Static maps that include walls and columns of buildings are relatively easy to construct. However, there are objects in the environment whose locations/states can change, such as doors, chairs, fences, stools for small children, and guide boards. Also, museum displays occasionally change, and some displays are actually movable. Such objects can be avoided after they become visible to robot sensors by re-planning paths. However, when a robot becomes aware of typical changes in the environment,